

PHYS505 Recitation — Feb 16, 2026

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Problem 2.2

Using the method of images for a point charge q inside a hollow grounded conducting sphere of inner radius a , find

- (a) the potential inside the sphere,
- (b) the induced surface-charge density,
- (c) the force acting on q .
- (d) What if the sphere is kept at potential V ? What if it has total charge Q on its inner and outer surfaces?

We choose the origin at the center of the sphere with the z -axis going through the position of the point charge (for azimuthal symmetry) so that its position is

$$r = z = R, \quad \vec{x} = R \hat{z}$$

The charge q' and its location z' of the image are then

$$q' = -\frac{a}{R} q, \quad z' = \frac{a^2}{R}, \quad \vec{x}_{q'} = \frac{a^2}{R} \hat{z}$$

- (a) The potential inside the sphere is a superposition of the potentials from the real charge inside and the image charge outside

$$\begin{aligned} \Phi(|\vec{x}| < a) &= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{|\vec{x} - R\hat{z}|} + \frac{q'}{|\vec{x} - z'\hat{z}|} \right) \\ \vec{x} - \vec{x}_q &= \vec{x} - R\hat{z} = \rho\hat{\rho} + z\hat{z} - R\hat{z} \\ &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\vec{x} - R\hat{z}|} - \frac{\frac{a}{R}}{|\vec{x} - \frac{a^2}{R}\hat{z}|} \right) \\ \Phi(\rho, z) &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z - R)^2}} - \frac{a}{R\sqrt{\rho^2 + (z - \frac{a^2}{R})^2}} \right) \end{aligned}$$

On the conductor's inner surface

$$\begin{aligned}\rho^2 + z^2 &= a^2 \\ \Phi(r=a) &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + z^2 + R^2 - 2Rz}} - \frac{a}{R\sqrt{\rho^2 + z^2 + \left(\frac{a^2}{R}\right)^2 - 2\frac{a^2}{R}z}} \right) \\ &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{a^2 + R^2 - 2Rz}} - \frac{1}{\sqrt{R^2 + a^2 - 2Rz}} \right) = 0\end{aligned}$$

satisfying the boundary condition. In the limit $R \rightarrow 0$, we get

$$\begin{aligned}\Phi(r < a) &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z-R)^2}} - \frac{1}{\sqrt{\frac{R^2}{a^2}(\rho^2 + z^2) + a^2 - 2Rz}} \right) \\ \Phi(r < a, R \rightarrow 0) &\rightarrow \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{a} \right)\end{aligned}$$

which is the potential for a point charge at the origin as expected, shifted by a constant to satisfy the boundary condition at $r = a$.

(b) We need the electric field for the surface-charge density

$$\begin{aligned}\vec{E} &= -\vec{\nabla}\Phi = -\frac{q}{4\pi\epsilon_0} (\hat{\rho}\partial_\rho + \hat{z}\partial_z) \left\{ \left[\rho^2 + (z-R)^2 \right]^{-1/2} - \frac{a}{R} \left[\rho^2 + \left(z - \frac{a^2}{R} \right)^2 \right]^{-1/2} \right\} \\ \frac{4\pi\epsilon_0}{q} \vec{E} &= \left\{ \frac{\rho}{\left[\rho^2 + (z-R)^2 \right]^{3/2}} - \frac{a}{R} \frac{\rho}{\left[\rho^2 + \left(z - \frac{a^2}{R} \right)^2 \right]^{3/2}} \right\} \hat{\rho} \\ &\quad + \left\{ \frac{z-R}{\left[\rho^2 + (z-R)^2 \right]^{3/2}} - \frac{a}{R} \frac{z - \frac{a^2}{R}}{\left[\rho^2 + \left(z - \frac{a^2}{R} \right)^2 \right]^{3/2}} \right\} \hat{z}\end{aligned}$$

The radial component is given by

$$E = E_r = \sqrt{E_\rho^2 + E_z^2}$$

On the surface, we have

$$\begin{aligned}\frac{4\pi\epsilon_0}{q} E_\rho(r=a) &= \frac{\rho}{[a^2 + R^2 - 2Rz]^{3/2}} - \frac{a}{R} \frac{\rho}{\left[a^2 + \left(\frac{a^2}{R} \right)^2 - 2\frac{a^2}{R}z \right]^{3/2}} \\ &= \left(1 - \frac{R^2}{a^2} \right) \frac{\rho}{[a^2 + R^2 - 2Rz]^{3/2}}\end{aligned}$$

for E_ρ and

$$\begin{aligned}\frac{4\pi\epsilon_0}{q}E_z(r=a) &= \frac{z-R}{[a^2+R^2-2Rz]^{3/2}} - \frac{a}{R} \frac{z-\frac{a^2}{R}}{\left[a^2+\left(\frac{a^2}{R}\right)^2-2\frac{a^2}{R}z\right]^{3/2}} \\ &= \left(1-\frac{R^2}{a^2}\right) \frac{z}{[a^2+R^2-2Rz]^{3/2}}\end{aligned}$$

for E_z . These have the correct limits

$$\begin{aligned}\theta=0, \pi \rightarrow \rho=0, z=\pm a, E_\rho &= 0 \\ \theta=\frac{\pi}{2} \rightarrow \rho=a, z=0, E_z &= 0\end{aligned}$$

Combining the components

$$\frac{4\pi\epsilon_0}{q}E_r(r=a) = \frac{\left[1-\left(\frac{R}{a}\right)^2\right]a}{[a^2+R^2-2Rz]^{3/2}}$$

Using $z = a \cos \theta$,

$$E_r(r=a, \theta) = \frac{q}{4\pi\epsilon_0 a^2} \frac{1-\left(\frac{R}{a}\right)^2}{\left[1+\left(\frac{R}{a}\right)^2-2\frac{R}{a}\cos\theta\right]^{3/2}}$$

which also has the correct limit for $R \rightarrow 0$. We can now find the charge density

$$\begin{aligned}\frac{\sigma}{\epsilon_0} &= \hat{n} \cdot \vec{E}(r=a) = -E_r(r=a) \\ \sigma(\theta) &= -\frac{q}{4\pi a^2} \frac{1-\left(\frac{R}{a}\right)^2}{\left[1+\left(\frac{R}{a}\right)^2-2\frac{R}{a}\cos\theta\right]^{3/2}}\end{aligned}$$

which is -ve everywhere on the surface. Integrating this, with $\alpha = R/a$, we find the total charge on the surface

$$\begin{aligned}\int \sigma dA &= a^2 \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta \sigma(\theta) = -\frac{q}{2} (1-\alpha^2) \int_{-1}^1 \frac{d\cos\theta}{(1+\alpha^2-2\alpha\cos\theta)^{3/2}} \\ &= -\frac{q}{2} \frac{1-\alpha^2}{2\alpha} \int_{-2\alpha}^{2\alpha} \frac{du}{(1+\alpha^2+u)^{3/2}} = -\frac{q}{2} \frac{1-\alpha^2}{2\alpha} \left. \frac{(1+\alpha^2+u)^{-1/2}}{(-1/2)} \right|_{-2\alpha}^{2\alpha} \\ &= q \frac{1-\alpha^2}{2\alpha} \left[\frac{1}{\sqrt{1+\alpha^2+2\alpha}} - \frac{1}{\sqrt{1+\alpha^2-2\alpha}} \right] = \frac{q}{2\alpha} (1-\alpha)(1+\alpha) \left(\frac{1}{1+\alpha} - \frac{1}{1-\alpha} \right) \\ &= \frac{q}{2\alpha} [(1-\alpha) - (1+\alpha)] = -q\end{aligned}$$

Why is this not q' ? Consider the outer region $r > b$ where b is the outer radius of the conducting shell. The potential on the outer surface $r = b$ is 0, so the

solution is $\Phi(r > b) = 0$. If we wrap a Gaussian sphere around the conducting shell at any $r > b$, the electric field and therefore the flux is 0, so total charge inside must be 0. The circuit that the conductor is connected to provides the charge ($-q$ in this case) required to cancel the effect of the point charge inside the conductor and that is all on the inner surface $r = a$, there is no charge on the outer surface. If this were an isolated conductor with a fixed charge Q , there would be a charge $Q + q$ distributed uniformly on the outside.

(c) Using the image charge (easy way), we get

$$\begin{aligned}\vec{F}_q &= \frac{1}{4\pi\epsilon_0} \frac{q q'}{(z_q - z')^2} (-\hat{r}) = -\frac{1}{4\pi\epsilon_0} \frac{q \left(-\frac{a}{R}q\right)}{\left(R - \frac{a^2}{R}\right)^2} \hat{z} \\ &= \frac{q^2}{4\pi\epsilon_0 R^2} \frac{\left(\frac{a}{R}\right)}{\left[1 - \left(\frac{a}{R}\right)^2\right]^2} \hat{z} = \frac{q^2}{4\pi\epsilon_0 a^2} \frac{\alpha}{(1 - \alpha^2)^2} \hat{z}\end{aligned}$$

(d) If the sphere is kept at potential V instead of 0, we can add a point charge q'' at the origin such that

$$V = \frac{q''}{4\pi\epsilon_0 a}$$

so the potential becomes

$$\begin{aligned}\Phi_V(\vec{x}) &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\vec{x} - d\hat{z}|} - \frac{1}{|\vec{x} + d\hat{z}|} \right) + \frac{q''}{4\pi\epsilon_0 |\vec{x}|} \\ \Phi_V(\rho, z) &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z - d)^2}} - \frac{1}{\sqrt{\rho^2 + (z + d)^2}} \right) + \frac{V a}{\sqrt{\rho^2 + z^2}}\end{aligned}$$

resulting in additional radial field on the surface

$$E_r(r = a, \theta) = \frac{q}{4\pi\epsilon_0 a^2} \frac{1 - \left(\frac{d}{a}\right)^2}{\left[1 + \left(\frac{d}{a}\right)^2 - 2\frac{d}{a} \cos \theta\right]^{3/2}} + \frac{V}{a}$$

extra charge $-q''$ distributed uniformly on the surface

$$\sigma(\theta) = -\frac{q}{4\pi a^2} \frac{1 - \left(\frac{d}{a}\right)^2}{\left[1 + \left(\frac{d}{a}\right)^2 - 2\frac{d}{a} \cos \theta\right]^{3/2}} - \frac{\epsilon_0 V}{a}$$

and additional force on q from the charge at the origin

$$\begin{aligned}
\vec{F}_q &= \frac{q^2}{4\pi\epsilon_0 d^2} \frac{\left(\frac{a}{d}\right)}{\left[1 - \left(\frac{a}{d}\right)^2\right]^2} \hat{z} + \frac{q''q}{4\pi\epsilon_0 d^2} \hat{z} \\
&= \frac{q}{d^2} \left\{ \frac{q}{4\pi\epsilon_0} \frac{\left(\frac{a}{d}\right)}{\left[1 - \left(\frac{a}{d}\right)^2\right]^2} + Va \right\} \hat{z}
\end{aligned}$$

If the sphere has total charge Q (which means it is isolated), we start with the solution for the grounded case which has charge $-q$ on the inner surface of the sphere in electrostatic equilibrium. After disconnecting the sphere, the additional charge $Q + q$ will be distributed uniformly on the outside and the solution for the interior doesn't change.